Catch Point Rate: How Much of a Quarterback's Yardage Is Ball Flight?

Working paper. Data: nflverse play-by-play, regular seasons 2006 to Week 8 of 2026 (air yards are not recorded before 2006).

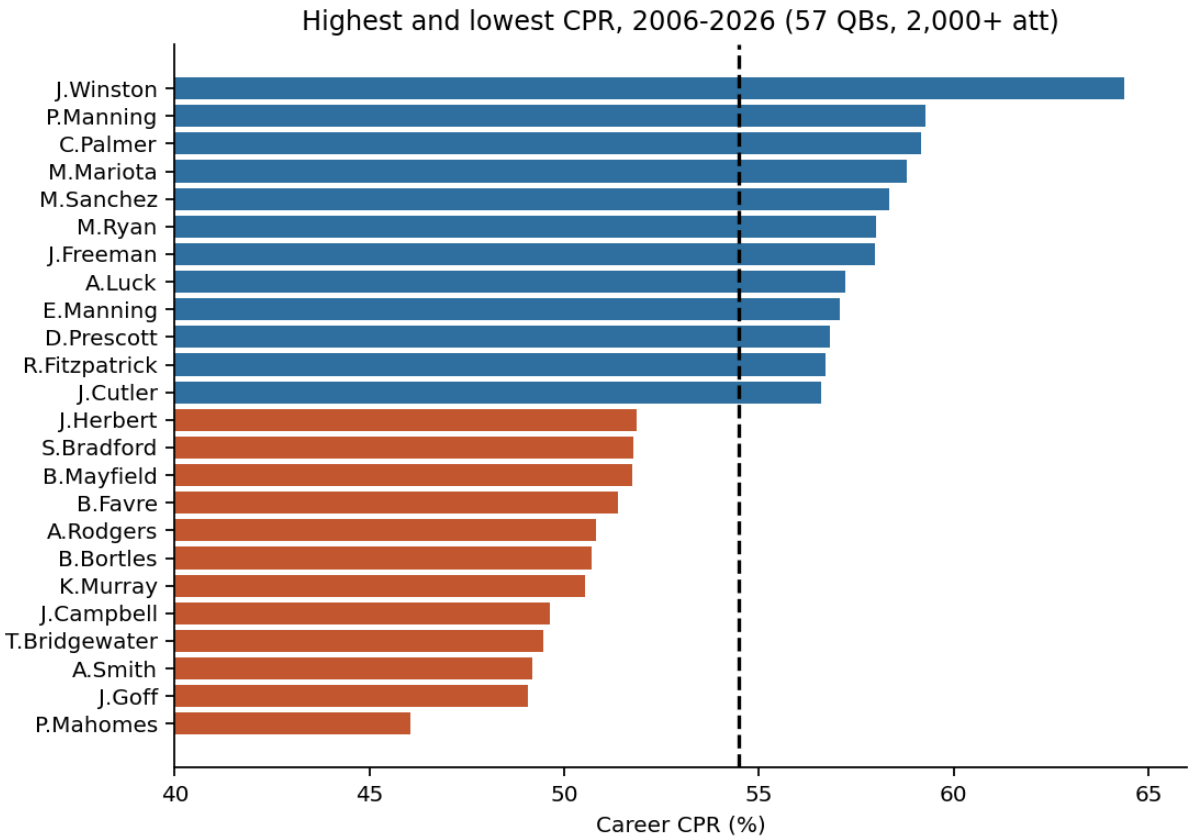
Abstract

We define **Catch Point Rate (CPR)**, the share of a quarterback's passing yards gained before the catch. Across 57 QBs with 2,000+ attempts since 2006, league CPR is 54.5%. CPR is strongly tied to average depth of target (r = 0.77), but the residual from that relationship isolates how much a QB's yardage depends on his receivers. League CPR has fallen from about 58% in 2006 to about 52% today, and Patrick Mahomes has the lowest career CPR in the sample (46.1%).

1. Definition

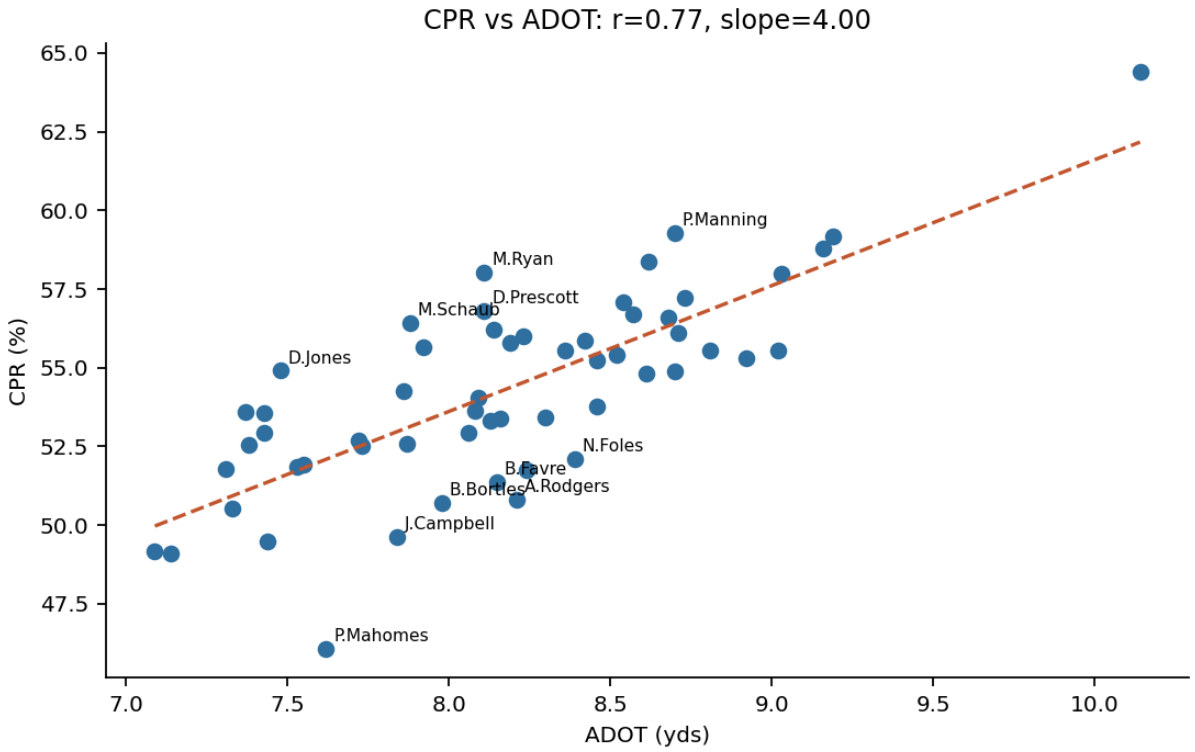
Using completions only, with sacks and spikes removed: **CPR = sum of air yards / sum of passing yards**. The remainder (1 minus CPR) is yards after the catch. Completed air yards can be negative on screens and dump-offs, which lowers CPR.

2. Career results



The dashed line is the league average. Jameis Winston (64.4%) is the clearest deep-ball outlier, followed by Peyton Manning (59.3%) and Carson Palmer (59.2%). At the other end, Mahomes (46.1%) sits nearly three points below the next QB, Jared Goff (49.1%).

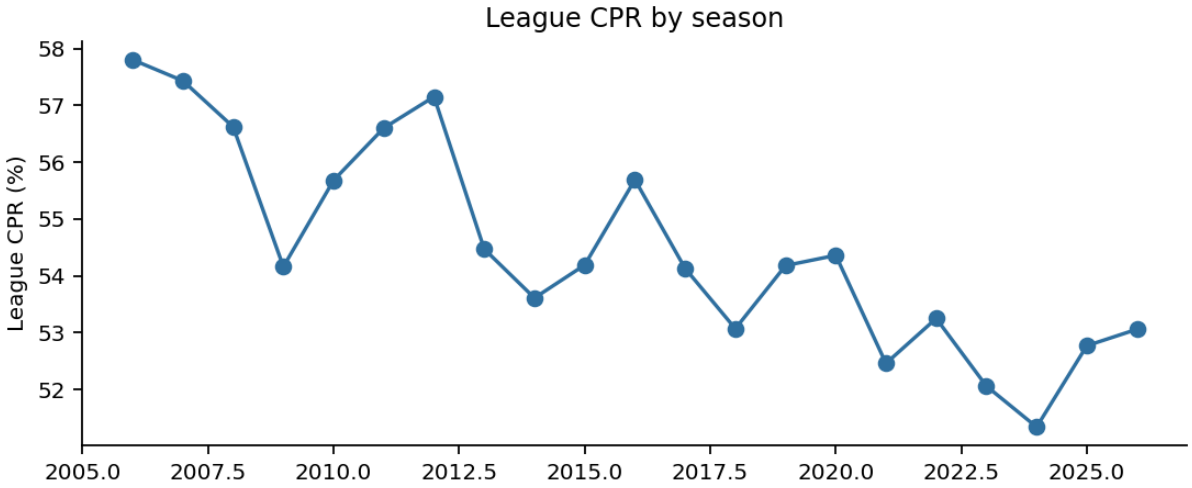
3. A model: CPR versus depth of target



Linear fit across the 57 QBs: CPR = a + 4.0 x ADOT, r = 0.77. Each extra yard of ADOT is associated with about 4 points of CPR. The **residual** (actual minus fitted) shows who gets more YAC than their depth predicts.

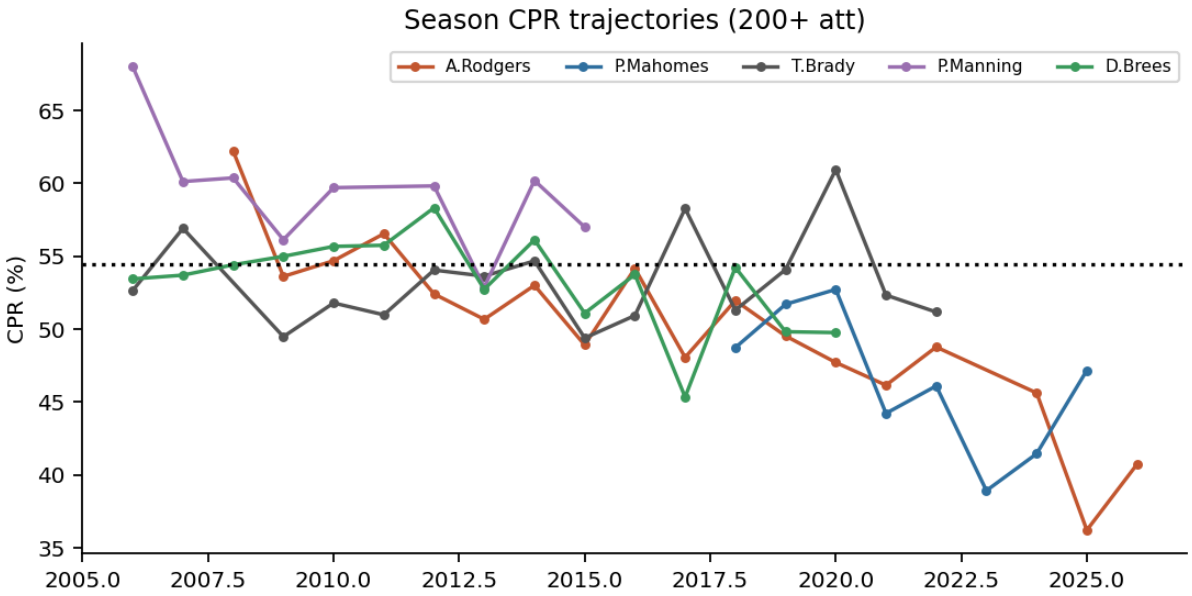
Most YAC-dependent (residual)		Most air-dependent (residual)	
Mahomes	-6.0	Ryan	+4.0
Rodgers	-3.6	D. Jones	+3.4
J. Campbell	-3.4	Schaub	+3.3
Foles	-3.1	P. Manning	+2.9
Favre	-2.8	Prescott	+2.8
Bortles	-2.8	Burrow	+2.5

4. League trend



League CPR peaked at 57.8% in 2006 and has trended down, hitting 51.3% in 2024. League ADOT fell over the same stretch (8.5 in 2006 to 7.7 in 2024), so the drop reflects a shift to shorter, higher-percentage throws, not a change in how receivers run after the catch.

5. Individual trajectories and stability



Season-to-season CPR for the same QB (300+ attempts in both years, 374 pairs) has a correlation of 0.46. That is moderate: CPR reflects a QB's tendencies, but scheme, receivers and age also move it. Extremes: Rodgers in 2025 (36.2%), Mahomes in 2023 (38.9%) and 2024 (41.5%) are among the lowest seasons since 2006 (alongside Garoppolo 2022 at 40.8% and Tagovailoa 2024 at 41.2%); Winston in 2018 (70.1%) and Bulger in 2007 (69.7%) are the highest.

6. Limitations

CPR cannot separate a receiver caught in stride from a play designed for YAC. It describes how the yards were produced, not how good the QB is. Pairing it with Next Gen Stats' YAC over expected would address both. Air-yard tracking is also less certain in the earliest seasons, and QBs with fewer than 2,000 attempts are excluded from the career table, so recent players are underrepresented.

7. Using it in an argument

A low CPR with a negative residual supports a scheme-driven claim (Rodgers, Campbell, Foles). A very low CPR next to elite results (Mahomes) is weaker evidence: his receivers may be earning the YAC, and the 2025 and earlier stats cannot tell those apart. Datasets: cpr_qb_career_2006_2026.csv, cpr_qb_seasons_2006_2026.csv, cpr_league_by_season_2006_2026.csv.